

The One-Graviton Universe: Simulating Spacetime Geometry as a Massless Spin-2 Field

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Graviton (hypothetical) | Spin: 2 | Mass: $< 10^{-22}$ eV | Force: gravity | Beyond SM **Source:** Extends v1.0Graviton Deep Dive Technical Report.pdf (V1.0)

Δ Change Log V1.0 → V2.0

Parameter	V1.0	V2.0	Source
LIGO O4 graviton mass bound	LIGO O1/O2 reference	V2.0: LIGO O4 (2023-2024): graviton mass $m_g < 1.27 \times 10^{-23}$ eV/c ² from gravitational wave dispersion. GW230529 binary system provides strongest individual event bound. In simulation: graviton phase velocity $v_{ph} = c \sqrt{1 - (m_g c^2/E)^2}$ deviates from c at cosmological scales	LIGO O4 2024; GW230529 paper
Gravitational wave polarization modes	GR predicts +,× modes only	V2.0: LIGO O4 polarization test: tensor (GR) modes dominant. Vector and scalar modes < 5% at 95% CL. Modified gravity (Brans-Dicke etc.) ruled out at tested parameter space. V2.0 simulation enforces spin-2 only: no scalar breathing mode	LIGO O4 polarization 2024

Parameter	V1.0	V2.0	Source
Quantum gravity footprints	General quantum gravity mention	V2.0: Fermion interference near black holes (Pikovski et al. 2024 theory): gravitational decoherence testable with next-generation GW detectors. Einstein Telescope (2030s). In simulation: graviton proximity to black hole adds quantum noise term to particle worldline	Pikovski 2024; ET Science Case
Massive gravity Vainshtein mechanism	Massless graviton assumed	V2.0: Vainshtein mechanism (nonlinear screening of massive gravity at $r < r_V = (r_g/m_g^4)^{1/5}$) allows massive graviton to mimic GR inside galaxies. $r_V \sim \text{Mpc}$ for $m_g \sim 10^{-33} \text{ eV}$. V2.0 implements: GR regime ($r < r_V$) vs. modified gravity ($r > r_V$)	Vainshtein 1972; Babichev & Deffayet review 2022

1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the “Origami Universe” metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,           // Spinor components
    berry_phase: f32,      // Geometric phase accumulated
    proper_time: f32,      // tau - worldline parameter
    color_charge: u32,      // Packed color in 2 bits (R=0,G=1,B=2)
```

```

    quantum_numbers: u32, // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
    entanglement_id: i32, // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step

    // Update worldline parameter
    p.proper_time += dtau;

    // Berry phase accumulation (topological phase)
    p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

    // Lorentz force (for charged particles)
    let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
    p.momentum.xyz += F * uniforms.dt;
    p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

    particles[i] = p;
}

```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m _e	9.10938356×10 ⁻³¹ kg	9.1093837139×10 ⁻³¹ kg
e	1.60217663×10 ⁻¹⁹ C	1.602176634×10 ⁻¹⁹ C (exact)
ħ	1.054571817×10 ⁻³⁴ J·s	1.054571817×10 ⁻³⁴ J·s (exact)
c	2.99792458×10 ⁸ m/s	2.99792458×10 ⁸ m/s (exact)
α	1/137.036	1/137.035999084 ±0.000000021

2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```

// V2.0 Universal Particle State – extends V1.0
struct ParticleInstanceV2 {
    // Inherited from V1.0
    double proper_time; // tau worldline parameter (meter)
    glm::dvec3 position; // 3-space vector
    glm::dvec3 momentum; // 3-momentum (relativistic)
    float spin_z; // +0.5 or -0.5 (or eigenvalue)
    float berry_phase; // Accumulated Berry/Aharonov-Bohm phase
}

```

